higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

ELECTROTECHNICS N5

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This marking guideline consists of 6 pages.

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QUESTION 1

1.1 By using interpoles (1)

1.2 1.2.1 Armature amp-turns $= (½) \times I_a/c \times Z/2p$
$= ½ \times 155/6 \times 800/6$✓
$= 1722,222$ AT(s)✓

$\theta = M° \times p = 6 \times 3 = 18$ E° ✓

Demag. amp-turns $= (½) x\ I_a/c \times Z/2p \times 4\theta/360$
$= 1722,222 \times (4 \times 18)/360$✓
$= 1722,222 \times 0,2$
$= 344,444$ AT(s) ✓

Cross-mag. amp-turns $= (½) \times I_a/c \times Z/2p \times [(1 - 4\theta)/360]$
$= 1722,222 \times (1 - 0,2)$ ✓
$= 1377,778$ AT(s)✓ (7)

1.2.2 $I_{Demag} = 344,444/1340$✓
$= 0,257$ A✓ (2)

1.3 1.3.1 Suddenly reduced: $N_2 = N_1$
$E_1 = V - I_{a1} R_a$
$= 120 - (58 \times 0,23)$
$= 120 - 13,34$
$= 106,66$ V✓

$E_2/E_1 = \phi_2 N_2/\phi_1 N_1$
$E_2/106,66 = 0,018/0,05$✓
$E_2 = 38,398$ V✓

$E_2 = V - I_{a2} R_a$
$38,398 = 120 - 0,23\ I_{a2}$✓
$I_{a2} = 81,7/0,23 = 355,217$ A ✓ (5)

1.3.2 Constant torque: $T_2 = T_1$
$T_2 / T_1 = \phi_2 I_{a2}/\phi_1 I_{a1}$
$1 = 0,018\ I_{a2}/(0,05 \times 58)$ ✓
$I_{a2} = 161,111$ A✓

$E_2 = V - I_{a2} R_a$
$= 120 - (161,111 \times 0,23)$
$= 120 - 37,056$
$= 82,944$ V✓

$E_2/E_1 = \phi_2 N_2/\phi_1 N_1$
$82,944/106,66 = 0,018\ N_2/(0,05 \times 750)$✓
$N_2 = 1620,844$ rev/min✓ (5)

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QUESTION 2

2.1 2.1.1 $X_C = X_L = 2\pi \times 50 \times 0,6 = 188,496\ \Omega$✓
$X_C = (2\pi fC)^{-1}$
$188,496 = (2\pi \times 50 \times C)^{-1}$✓
$C = (2\pi \times 50 \times 188,496)^{-1}$✓
$= 1,689 \times 10^{-5}$ F
$= 16,887\ \mu F$ ✓

OR

$F_r = (2\pi \sqrt{LC})^{-1}$
$50 = (2\pi \sqrt{0,6\ x\ C})^{-1}$✓
$\sqrt{0,6\,x\ C} = (2\pi \times 50)^{-1}$ ✓
$C = (2\pi \times 50)^{-2}/0,6$ ✓ = 16,887 µF✓ (4)

2.1.2 $Z_T = R = 8\ \Omega$
$I_T = V_T/Z_T = 125/8 = 15,625$ A✓

$V_R = 15,625 \times 8 = 125$ V✓
$V_L = V_C = 15,625 \times 188,496 = 2945,25$ V✓ (3)

2.2 2.2.1 $\cos\Theta = P/VI = 2100/(240 \times 10)$✓ = 0,875✓
$\Theta = \cos^{-1}(0,875) = 28,96°$✓

$Z_1 = V_1/I_1 = 240 < 0/10 < -28,96 = 24 < 28,96\ \Omega$✓

$I_T = 26 < -29,54 = 22,62 - j12,819$
$I_1 = 10 < -28,96 = 8,749 - j4,842$
$I_2 = I_T - I_1 = 22,62 - j12,819 - 8,749 + j4,842$✓
$= 13,871 - j7,977$
$= 16,001 < -29,9$ A ✓

$Z_2 = 240 < 0/16,001 < -29,9 = 14,999 < 29,9\ \Omega$✓

$Z_{SE} = Z_1 + Z_2$
$= 20,999 + j11,621 + 13,003 + j7,477$✓
$= 34,002 + j19,098$
$= 38,998 < 29,32\ \Omega$✓

$I_{SE} = V/Z_{SE} = 240 < 0/38,998 < 29,,32$✓
$= 6,154 < -29,32$ A✓ (11)

2.2.2 $\cos\Theta = \cos 29,32$ ✓ = 0,872 lagging ✓ (2)
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QUESTION 3

3.1 At no load the power factor is very low and lagging whereas the power factor at load is nearly equal to the power factor of the load which the transformer is carrying. (2)

3.2
- Making a transformer window long and narrower
- Arranging the primary and secondary windings concentrically
- Sandwiching the primary and secondary windings
- Using a shell-type construction (3)

3.3 3.3.1 $R_{E1} = 0,2 + 0,03(4)^2$✓ = 0,68 Ω✓

$X_{E1} = 1,4 + 0,045(4)^2$✓ = 2,12 Ω✓

Reg = 67,5[(0,68 × 0,86) + (2,12 × Sin30,68)]✓/2000
= 112,487 /2000 ✓
= 0,056 p.u = 5,624 % ✓

$V_{2FL} = V_{2NL} – Reg \times V_{2NL}$
= 500 – (0,056 × 500)✓
= 500 – 28 = 472 V✓ (9)

3.3.2 %X = (67,5 × 2,12)✓/2000 = 0,072 × 100 = 7,155%✓ (2)

3.3.3 %R = (67,5 × 0,68)✓/2000 = 0,0229 × 100 = 2,295%✓ (2)

3.3.4 %Z = 2,295 + j7,155 ✓ = 7,514 < 72,22 Ω✓ (2)
[20]

QUESTION 4

4.1 4.1.1 $P_{IN} = \sqrt{3} V_L I_L \cos \Theta$
$I_L = P_{IN}/(\sqrt{3} V_L \cos \Theta) = 82 \times 10^3 /(\sqrt{3} \times 2200 \times 0,899)$✓
= 23,937 A ✓
$I_P = I_L/\sqrt{3}$ = 23,937/3 = 13,82 A✓ (3)

4.1.2 $S = \sqrt{3} V_L I_L = \sqrt{3} \times 2\,200 \times 23,937$ = 91,212 kVA✓ (1)

4.2 4.2.1 $d_e = \sqrt[3]{40 \ x \ 40 \ x \ 80}$ = 50,397 cm✓

L = 10[0,05 + 0,2 ln(50,397/0,4)] mH✓
= 10(0,05 + 0,967)
= 10,172 mH✓

C = 10[18 ln(50,397/0,4)]$^{-1}$✓
= 10(0,01149)
= 0,11487 µF✓ (5)

4.2.2 $D_e = \sqrt[3]{40 \ x \ 70 \ x \ 100} = 65,421$ cm ✓

$L = 10[\ 0,05 + 0,2 \ln(65,421/0,4]$ mH✓
$= 10(\ 0,05 + 1,019\ )$
$= 10,694$ mH ✓

$C = 10[\ 18 \ln(65,421/0,4)\ ]^{-1}$ ✓
$= 0,109$ µF✓ (5)

4.3 $\text{Tan}\Theta = \sqrt{3}(W_2 - W_1)/(W_2 + W_1) = \sqrt{3}(850 - 400)/(850 + 400) = 0,634$✓
$\Theta = \text{Tan}^{-1}(0,6234) = 31,96^0$ ✓

$I_L = P_T / (\sqrt{3}\ V_L \text{Cos}\Theta) = 1250 / (\sqrt{3} \times 120 \times \text{Cos } 31,96)$✓ $= 7,089$ A✓

$I_P = I_L / \sqrt{3} = 7,089 / \sqrt{3} = 4,093$ A✓

$Z_P = V_P / I_P = 120/4,093 = 29,318$ A✓ (6)
[20]

QUESTION 5

5.1
- Electromechanical counter
- Mechanical differential counter
- Stroboscopic method (Any 2 × 1) (2)

5.2 $V_P = V_L / \sqrt{3} = 12 \times 10^3 / \sqrt{3} = 6928,203$ V✓

$f = N_S\ p/60$
$p = 60f/N_S = (60 \times 50)/600 = 5$ pole pairs✓
Poles = 10✓

$E_P = 2,22\ K_P\ K_d\ Zf\phi$
$Z_P = E_P/(2,22\ K_P\ K_d\ f\phi)$
$= 6928,203/(2,22 \times 0,97 \times 50 \times 0,18)$✓
$= 357,481$ conductors ✓

$S_{PH} = 6 \times 10 = 60$ slots/ph✓

Z/S = 357,481/60 = 6 cond/slots✓ (7)

5.3 5.3.1 $E_2 = E_L / \sqrt{3} = 170 / \sqrt{3} = 98,149$ V✓
$E_R = SE_2 = 0,04 \times 98,149 = 3,926$ V✓

$Z_R = R_2 + j(SX_2) = 0,8 + j(0,04 \times 4,2)$✓
$= 0,8 + 0,168 = 0,817 < 11,86\ \Omega$✓

$I_R = E_R / Z_R = 3,926/0,817 = 4,805$ A✓ (5)

5.3.2 $S = R_O / X_O = 0,8/4,2$✓ $= 0,19 \times 100 = 19,048\ \%$✓

$E_R = SE_O = 0,19 \times 98,149 = 18,648$ V✓

$X_2 = SX_2 = 0,19 \times 4,2 = 0,798\ \Omega$✓

$Z_R = 0,8 + j0,798 = 1,129 < 44,93\ \Omega$✓

$I_R = E_R/Z_R = 18,648/1,129 = 16,517$ A✓ (6)
[20]

TOTAL: 100

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